

Lesson Objective

Interpret signed numbers when used to represent time in situations about speed and direction as well as understand that the product of two negative numbers is positive.

1. Multiply a positive number by a positive number and a positive number by a negative number to review known sign rules. Evaluate each product. a) $6 \cdot 2$ b) $6 \cdot (-2)$ <i>Answer: a) 12; b) -12</i>	Ask students to evaluate $6 \cdot 2$ and share how they know the product is positive. Ask students to evaluate $6 \cdot (-2)$. Prompt them to think of this as 6 groups of -2, landing at -12 on a number line. Ask students to name the sign rule each product shows: a positive times a positive is positive, and a positive times a negative is negative.
2. Use the distributive property to discover that the product of two negative numbers is positive. Evaluate the expression $-4 \cdot (7 + (-2))$ two ways. First, evaluate what is inside the parentheses, then multiply. Then, use the distributive property to rewrite the expression as a sum of two products. Use what you know about the first product and the total to figure out the value of the second product.	Tell students you will evaluate this expression two different ways and the answer should match. Guide students through evaluating inside the parentheses first: $7 + (-2) = 5$, then $-4 \cdot 5 = -20$. Record this answer. Now guide students to use the distributive property. Write $-4 \cdot 7 + -4 \cdot (-2)$. Ask students to evaluate $-4 \cdot 7$ to get -28. Point out that -28 plus the second product must equal -20.

Answer: $-4 \cdot (7 + (-2)) = -4 \cdot 5 = -20$; using the distributive property, $-4 \cdot 7 + -4 \cdot (-2) = -28 + -4 \cdot (-2) = -20$, so $-4 \cdot (-2) = 8$.	Ask students what number added to -28 gives -20. Guide them to see that the second product must be 8, so $-4 \cdot (-2) = 8$. Tell students that for the distributive property to work for all rational numbers, a negative times a negative must be positive.
3. Apply the rules for multiplying signed numbers to evaluate products involving negative factors. Evaluate each product. a) $-5 \cdot (-3)$ b) $-7 \cdot (-4)$ c) $-6 \cdot 4$ Answer: a) 15; b) 28; c) -24	Students work independently. After students finish, ask them to name the sign rule used for each product.

Monitor For	Instructional Moves
• In Problem 1, do students correctly identify that $6 \cdot (-2)$ is -12 by thinking of it as 6 groups of -2? • In Problem 2, do students correctly evaluate $7 + (-2)$ as positive 5, or do they get tripped up adding a positive and a negative integer? • In Problem 2, do students reason that $-28 + 8 = -20$ to conclude that $-4 \cdot (-2) = 8$, or do they guess the sign without checking the sum?	• In Problem 1, reinforce the connection to repeated addition by asking students to say "6 groups of -2" before computing $6 \cdot (-2)$ so they anchor the sign in the meaning of multiplication. • If a student struggles to add $-28 + 8$ in Problem 2, sketch a number line and ask them to start at -28 and move 8 to the right to land at -20.

• In Problem 3, do students apply the rule that a negative times a negative is positive, or do they default to a negative product?	• In Problem 2, emphasize the conceptual reasoning: state that because the distributive property must hold for all numbers, $-4 \cdot (-2)$ has to equal positive 8 for both methods to give the same result. • If a student writes a negative product in Problem 3, redirect them to Problem 2 and ask what sign two negative factors produce.
Reflection Questions • What sign do you get when you multiply two negative numbers? How do you know? <i>Answer: A positive. We saw in Problem 2 that $-4 \cdot (-2)$ had to equal positive 8 for the distributive property to give the same answer as evaluating inside the parentheses.</i> • How did evaluating $-4 \cdot (7 + (-2))$ two different ways help you figure out the value of $-4 \cdot (-2)$? <i>Answer: Both methods have to give the same answer, -20. Since $-4 \cdot 7 = -28$, the second product had to be 8 to make the sum equal -20.</i> • How would you evaluate $-7 \cdot (-4)$ and explain the sign of the product? <i>Answer: $7 \cdot 4 = 28$, and a negative times a negative is positive, so $-7 \cdot (-4) = 28$.</i> • How is multiplying a negative by a negative different from multiplying a positive by a negative? <i>Answer: A positive times a negative gives a negative product, like $6 \cdot (-2) = -12$. A negative times a negative gives a positive product, like $-5 \cdot (-3) = 15$.</i>	